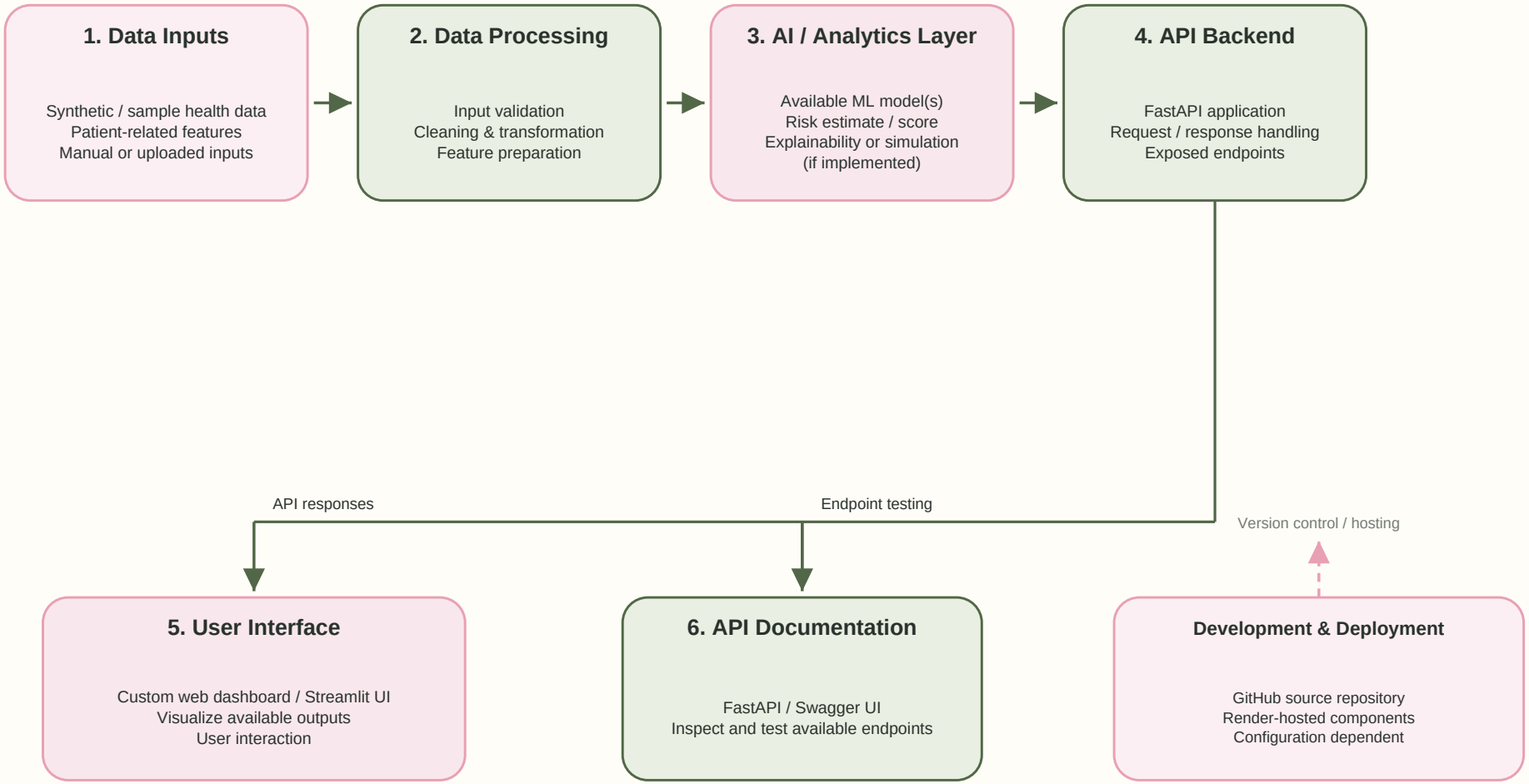
VitaTwin AI — System Architecture

Predict. Simulate. Prevent. | Research & educational prototype



Prototype scope Conditional components must be verified against the current codebase.

Not a medical device • Not for diagnosis or treatment

Architecture Notes & Data Flow

Implementation-aware summary for the Digital Twin Challenge submission

Request Flow

1. A user provides or selects supported input data.
2. The interface sends a request to the backend, where supported.
3. The backend validates the request and invokes available logic.
4. The response is returned for display or inspection.

Data & Model Boundaries

Use only data sources actually connected to the current build.
Synthetic/sample data must be labelled as such.
Model outputs are estimates, not clinical conclusions.
Do not report metrics unless experimentally measured.

Security & Privacy Considerations

For any future real-patient deployment: obtain appropriate consent, minimize collected data, protect secrets, add authentication and role-based access, encrypt data in transit and at rest, define retention policies, and follow applicable requirements.

Validation & Future Work

Verify model integration, input schemas, endpoint behaviour, and frontend/API connectivity. Future clinical use would require external validation, fairness and robustness testing, clinical oversight, usability assessment, and regulatory review where applicable.

High-level diagram: confirm every component against the deployed implementation before submission.